

Research Proposal

In this section, you must present your proposal by developing the following aspects: a) Theoretical-conceptual foundations and state of the art that support the proposal, b) Hypothesis or research questions and objectives, c) Scientific or technological novelty of your proposal, d) Methodology, e) Work plan or Gantt chart and f) Background information to assess the capacity of the team to implement the proposal.

Remember that:

- You must strictly comply with what is established in Bases Concurso Nacional de Proyectos Fondecyt Regular 2024.
- All text, paragraphs or textual phrases from a bibliographic reference, whether by other authors or their own, must be duly identified in the text and in the list of references.

The file must contain a maximum of **10 pages** (use letter size format, Verdana font size 10 or similar).

Aspects related to the proposed research that are included in annexes will not be considered in the evaluation process.

Introduction

Modeling scientific and engineering processes using numerical methods, and rendering models or physical phenomena in computer graphics, among others, usually require an adequate spatial discretization (mesh) of a domain or object. Hydrological modeling, device simulation, fluid dynamics, fracture mechanics, or real time visualization considering physics, among others, are some applications. Depending on the problem to be solved and the numerical method to be used, mesh cells have to fulfil specific geometric properties and the whole mesh must have a proper point distribution. These issues make the problem of generating good quality meshes with the minimum amount of points still very challenging, in particular for the 3D space.

One of the most studied spatial discretization is the Voronoi diagram, because it helps to solve many geometric problems in nature like rock packing, strategic object placements and patterns of growth in crystal simulations, among others. The Voronoi diagram of a point set P is composed of convex polytopes (Voronoi regions) where each Voronoi region contains a seed point p in its interior and it is surrounded by all the points of the space that are closer to p than any other point q , with $p, q \in P$. This definition extends to n -dimensional space. Its dual in 2D(3D), the Delaunay partition of the point set P is a triangulation (tetrahedral mesh) if there are no co-circular(co-spherical) points. This triangulation (tetrahedral mesh) fulfills the Delaunay condition, i.e., the circumcircle(circumsphere) of each mesh cell does not include any other point $p \in P$ in its interior. Among the good properties of a 2D Delaunay partition is that it maximizes the size of its smallest angle over all the possible partitions of a point set P . In 3D this property does not hold.

Because of the geometric properties of the Voronoi diagram and of the Delaunay partition, and the availability of robust and open source tools to generate them, both discretizations are very often used together with numerical methods that accept triangles/tetrahedra or convex polygons/polyhedra as basic cells. This is the case for some problems solved with the Finite Element Method (FEM) where proper triangulations generated by Detri2 [1] or Triangle [2] in 2D or Tetgen in 3D [3] are perfect. In case problems solved with the Polygonal Finite Element Method (PFEM), a mesh composed of convex polygons/polyhedra with the required point distribution can be appropriate and so constrained Voronoi Meshes [4] are usually chosen. In the last two decades, the Virtual Element Method (VEM) appears, convex/non-convex of any shape can be used [5, 6], and usually meshes based in the Voronoi diagram such as the generated by PolyMesher [7] are used. The recent works published in [8, 9] make a key contribution studying the role of the cell shapes to achieve good convergence results. If problems are solved with the Control Volume Method(CVM), the combination of both, the Delaunay partition and the Voronoi diagram is an option. Proper mixed element Delaunay meshes for the CVM are described in [10]. Together with tetrahedra, the mesh includes cuboids, prism and pyramids among others while these elements are co-spherical.

Since the last eight years we have been working in the design and improvement of meshing algorithms to generate polygonal and polyhedral meshes for diverse applications. In 2D, we have developed (i) polygonal meshes inspired in packing algorithms [11], (ii) polygonal meshes representing catchments for surface hydrological modeling [12, 13], (iii) 2D constrained Voronoi meshes and a VEM library (Delanoi and Veamy tools, respectively) [14], (iv) 2D polygonal meshes built from triangulations both designing sequential and parallel algorithms (Polylla [15] and GPolylla [16] tools), and in compact space [17]. In 3D, our progress has been slower than in 2D. However we can mention (i) the integration of a polyhedral Delaunay tessellation strategy [18] inside of a modified octree meshing tool [19, 20], (ii) applying a Delaunay

partition for searching cosmic polyhedral voids [21] and (iii) preliminary design and implementation of polyhedral meshes from tetrahedralizations. This research proposal is presented as a logical continuation of my current Fondecyt Project but with emphasis in 3D polyhedral meshes for the VEM. That is why part of this proposal is a revisited and updated version of the hypotheses and goals of my current project considering what has already worked and what has to be modified. In addition, we want to orient part of the efforts of this project to extend our previous work for hydrological modeling in catchments [12, 13] to incorporate the challenges coming from modeling aquifers with polyhedral meshes and so the interchange of water between both components. To count with this kind of simulations is extremely important in our country nowadays to estimate the water resources available for the community and agricultural use because of the climate change.

In my current Fondecyt Project, we have developed an efficient and robust polygonal mesh generator [15] that starting from a constrained Delaunay triangulation, with the required density distribution, generate a proper polygonal mesh for the VEM. Since the boundary is already fitted by the triangles and the algorithm builds polygons joining triangles according to some criteria, the final polygons respect the boundary/interfaces of the domain. The main difference with meshes based on the Voronoi diagram is that the initial cells are from the beginning inside the domain and polygons might be non-convex. In addition, as we expected, this kind of polygonal mesh can fit the input geometry and point distribution using less points and polygons than a Voronoi constrained mesh [15]. Currently, we are adding some strategies for automatic refinement and optimization of the mesh, but this step will not be ready in the current project. That is why this proposal will include further improvements in 2D as well.

Automatic algorithms to automatically generate generic and appropriate polyhedral meshes that at the same time take care of reducing as much as possible the number of points and cells, have not been developed yet [9]. As mentioned before, the most used meshes are meshes based on the Voronoi diagram because there are several algorithms and tools to generate and adapt them to the VEM requirements, but as mentioned in [9], there is still space to develop new algorithms that are more flexible and efficient than the current ones. In addition, we believe that the requirements of each specific numerical method and application can be properly encapsulated so that we can design meshing algorithms to develop a flexible mesh generator easily adaptable to new requirements.

This proposal is organized as follows: it starts with the motivation of this research project, continues with the challenges in the meshing field and in particular for polygonal and polyhedral meshes, and with our approach included as part of the conceptual framework and research problem. Then, we describe the related work, and present our hypotheses, the general and specific goals, the methodology and work plan, to finally end with a summary of the work in progress.

Motivation

The Virtual Element Method (VEM) [22, 23] can be used with cells of arbitrary shape. The mathematical expressions for the linear system can be analytically solved; then the solutions are free of integration errors. The VEM has been used to simulate several problems in a more precise and efficient way than the PFEM and standard numerical methods [24]. In this section we describe the scope of this proposal in the sense of which kind of applications are motivating our research to design scalable and robust algorithms for the generation of polyhedral meshes.

- Numerical solutions for physical phenomena on complex domains using the VEM. The interest to count with automatic algorithms to generate meshes composed of generic polyhedral with the required quality is rapidly increasing [9]. One of these complex phenomena is to model fractures produced in underground rock formation. In particular, for fracture simulation, a good quality mesh must have regions with different levels of refinement according to what the simulation requires. Several mesh quality indicators have been proposed to ensure the VEM convergence and low approximation error but recent studies have shown that these have been overestimated [8]. In 2D, it is recommended that the non-convex polygonal cells are star-shaped and the length of each edge of the polygonal cell must be larger than a certain value, but in several experiments applying the VEM over meshes that break these criteria might converge too [8]. Avoiding too small or large angles, or cells with bad aspect ratio, as geometric features required for triangles or quads in FEM are not needed for polygonal cells. It may be necessary to locally refine a coarse cell into polygons/polyhedra, for example modeling fractures, if the fracture goes through and add new points close to the crack tip in order to compute the stresses around it [25]. The transition between fine and coarse regions is also recommended. Therefore automatic and efficient algorithms to refine and optimize polygonal and polyhedral cells to fulfil the quality requirements are needed too.
- Simulation of hydrological processes: The requirements come from the numerical models used to simulate hydrological processes at urban and peri-urban scales. The initial mesh is obtained by intersecting polyline layers

of land use, rivers, or sewers among other natural or relevant man-made features. Initial meshes composed of usually non convex polygons very bad shaped with holes must be improved in order to generate proper polygons, keeping low the number of points and polygons while preserving the main geometric features. Some geometric constraints for the mesh cells are: avoid too elongated elements, small internal angles and very abrupt area changes, between neighboring cells. Moreover the edge that connect the centroid of two adjacent cells must be orthogonal (or pseudo-orthogonal) with respect to the shared edge/face (common interface). Our approach in 2D was to generate a triangulation that exactly fits the geometry (initial mesh), and then join triangles to form adequate convex/non-convex polygons [12, 13]. In 3D, octree based meshes with the control volume method have been used to model groundwaters [26]. Currently we are exploring how to mix superficial hydrological and groundwater models to study the water resources in Chile. As a consequence of the change to the Water Code in Chile (as of April 2022) it is now a legal obligation to carry out surface and groundwater modeling in Chile's watersheds. Therefore, the improvement of aquifer modeling meshes would allow us to be aligned with the new requirements of the water authority and to improve main tools oriented to support the water management in Chile.

- Semiconductor device simulation: our group decided to approach this problem using mixed-element Delaunay meshes composed of cuboids, prisms, pyramids and tetrahedra and the Finite Volume Method(FVM)[27, 28]. We required both the edges of the Delaunay tessellation and the Voronoi regions as finite volumes in order to simplify the computation of the numerical integration. Our challenge was to automatically generate polyhedron Delaunay meshes for complex domains with the minimum number of points and cells that fulfil the point density requirements. As we will see below, these kind of meshes seems to be adequate for hydrological modeling too. In order to advance in this direction, in the last years we have integrated a strategy [18] that partition a convex element with Steiner points into co-spherical polyhedral elements into the mesh generator Mesher [19], a mixed element mesh generator based on modified octrees. Moreover we are allowing the user to control the transition between neighboring elements so that we can generate a set of related meshes to test how the shape of the polyhedral elements influence the convergence results in the VEM. We are currently evaluating the impact in the point and element reduction in comparison with previous approaches.

Challenges on polygonal and polyhedral meshes

There are several numerical methods, and each one has a range of applications where they give accurate results in a reasonable time, but there is still no method that can be used to solve any problem [29, 9]. In this sense, the VEM, a quite new method, is being explored because it is allowing researchers and engineers to solve complex problems in a robust and efficient way [29]. Several authors such that Wriggers et al. also state that the application range of VEM is still not known and so it is still an open challenge to determine it. Since the mesh can be formed by generic polygons/polyhedra, the research community expects that quality polygonal/polyhedral meshes are composed of a much lower number of mesh elements in comparison with the number of elements of meshes composed of simple convex elements. Therefore, the VEM should allow us to model large and difficult problems that are not possible to solve with standard numerical methods should be .

We can summarize the main challenges in the meshing field independent of the cell type as the need of more robust and efficient algorithms for the:

- Automatically generation of good quality meshes for complex domain geometries and size problems (hydrological, geological and climate modeling, among others)
- Efficient handling of huge meshes through the simulation pipeline in order to implement and execute more complex and accurate models than now (GPU-computing, multicore, clusters)

In the last years, several advances in the direction of the identification, design, and development of quality metrics for polygonal/polyhedral cells to guide the point insertion and mesh quality improvement process has been done [8, 30, 9]. As we mentioned before, several approaches have been used to generate polygonal/polyhedral meshes. These are (i) Voronoi meshes [31, 32, 33, 7, 24, 34], (ii) triangulations(tetrahedral meshes), whose triangles(tetrahedra) are divided into polygons (polyhedra) [35, 36] and (ii) our recent work building generic polygons by joining triangles [15]. These approaches still do not solve several challenges in the generation of polygon/polyhedron meshes, among which we can mention:

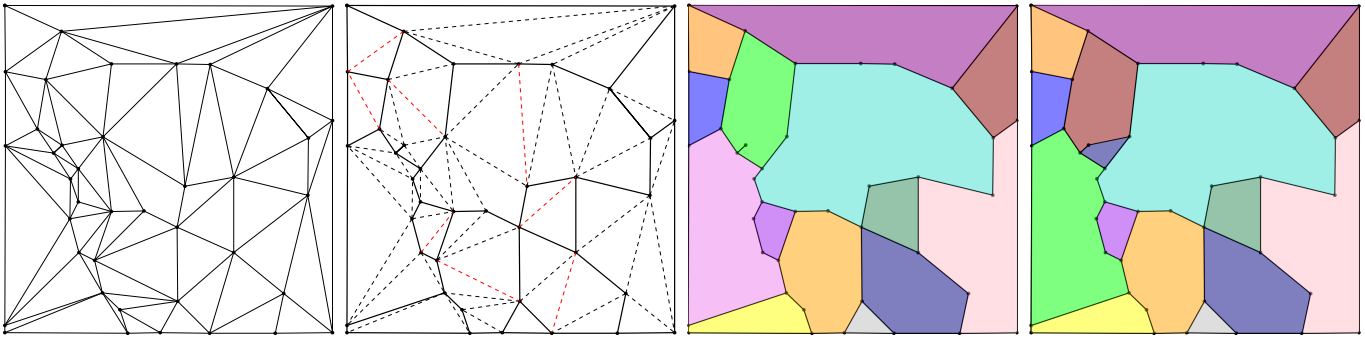


Figure 1: Geometric concepts: Delaunay triangulation of a point set(left), terminal-edges, frontier-edges and terminal-edge regions(middle left), terminal-edge regions(middle right), and polygonal mesh(right)

- The need of efficient and robust algorithms to generate good enough starting polyhedral meshes on complex domains using as less cells as possible.
- Design of strategies to efficiently divide polygons/polyhedra into new polyhedral cells hopefully by adding as few points as possible guided by quality metrics [8]
- Design of scalable polygon/polyhedral meshing algorithms through the development of parallel algorithms

Conceptual framework and research problem

As we have told in the previous section, there is still no numerical method that can be used to solve any problem. The same occurs with the kind of meshes. There is no a unique mesh type that can be used as a good discretization of geometric domain to solve any problem. In this section, we will first introduce the concept of terminal-edge region and how we have used it in our current research project to generate adequate polygonal meshes that are an alternative to constrained Voronoi meshes. Second, we will define the concept of terminal-edge and terminal-face regions in 3D and discuss if polyhedra could be built from them. Third, we are going to state several research questions and fourth the main research lines we want to explore to answer the previous questions. Finally, we are going to summarize the results of our ongoing research project and how this proposal will permit us to continue the research in improving algorithms for the generation of quality polyhedral meshes.

Terminal edge regions are bounded by edges that are not the longest edge of any triangle (frontier-edges). In case of isosceles or equilateral triangles, a random order is given to the edges [37, 38]. Terminal edge regions can also be seen as by the union of the triangles belonging to the longest-edge propagation path of the triangle t ($\text{Lepp}(t)$) [39, 40] that share the same terminal-edge. An edge is called terminal-edge when it is the longest edge of the triangles that share it. Figure 1 shows the terminal-edge regions built from a Delaunay triangulation of a point set (left) and the final polygonal mesh (right). Terminal-edges are shown with dashed red lines, frontier-edges with black solid lines, and interior edges (edges that are the longest edges of only one triangle of the two triangles sharing them) with black dashed lines at Figure 1(middle-left). Terminal-edge regions can generate non-simple polygons as it can be observed in the green polygon at Figure 1(middle-right). The non-simple polygons are transformed into simple polygons by adding one or more internal edges chosen from the initial triangulation according to some user criterion (see Figure 1 at the right). During the current project, we have further study the properties of terminal-edge regions, developed a polygonal mesh generator called Polylla, designed and run several experiments to study the kind of polygonal meshes that can be generated from terminal-edge regions. This kind of meshes were also validated by solving standard problems with the VEM and compared with constrained Voronoi meshes both in terms of element shapes and mesh size and accuracy of the simulation results. Figure 2 shows three meshes. The left one is a conforming Delaunay triangulation, the middle mesh is a constrained Voronoi mesh generated from the left Delaunay triangulation, and the right polygonal mesh is the one generated with Polylla. Both the conforming Delaunay triangulation and the constrained Voronoi mesh were generated with Detri2qt. In [15], we describe in detail the properties of terminal-edge regions and the developed algorithms. The main results can be summarized as follows: (i) These meshes contain in average three times less polygons and half the number of points than the standard constrained Voronoi meshes starting from the same input, (ii) the numerical results show that the accuracy computational cost of the VEM numerical solution using Polylla and Voronoi meshes are similar, (iii) Polylla meshes are simpler and faster to generate than Voronoi meshes starting from the same conforming input

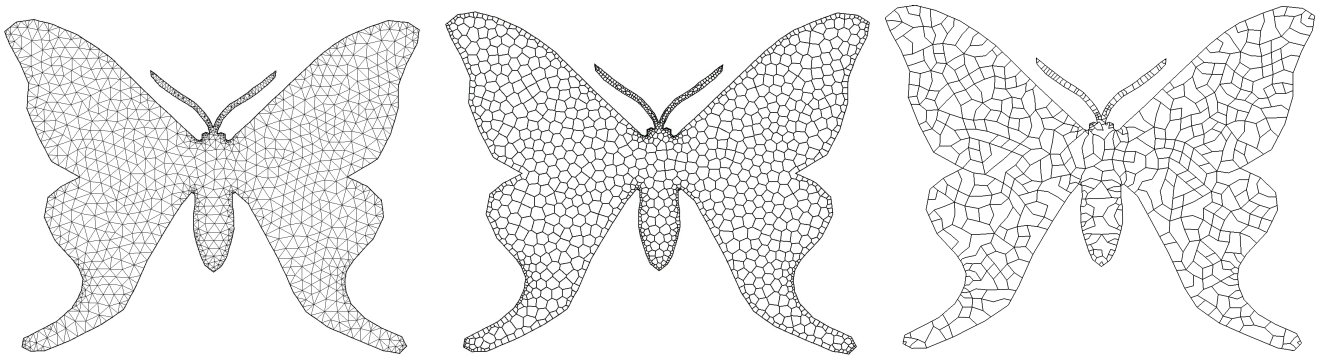


Figure 2: Three kind of meshes: Delaunay Triangulation (left), Constrained Voronoi mesh (middle), Polylla mesh (right)

triangulation. Polylla meshes respect the input point distribution; Voronoi based meshes use these points as seeds and insert Voronoi points, one per each input triangle. Since the number of triangles is larger than the number of input points, the Polylla mesh has less points and polygons than the Voronoi based mesh, and (iv) Polylla meshes can be generated from any constrained triangulation; it does not require a Delaunay triangulation as input. However we have observed a that starting from a Delaunay triangulation, the final polyonal mesh has less polygons with larger area than starting from any other triangulation.

In the current project we have validated our hypothesis related to terminal-edge regions and 2D polygonal meshes but it is still open the question if terminal-edge regions can be adapted to be good basic cells for polyhedron meshes? The concepts of terminal-edge and Lepp have been already extended for tetrahedra [39, 41] so the concept of 3D terminal-edge region can be seen as the union of the tetrahedra t whose $Lepp(t)$ (Longest-edge propagation path) share the same terminal-edge. The boundary of a 3D terminal-edge region is formed by frontier and barrier faces. We have done preliminary experiments to evaluate which kind of polyhedral elements can be generated from 3D terminal-edge regions and we have obtained many non-simple polyhedra, connected by points and edges. Final simple polyhedra are formed from two or three tetrahedra and that it is not what we have expected. In similar experiments we have observed that if the concept $Lepp(t)$ is extended to $Lfpp(t)$ (Longest-face propagation path) were "Longest" can be associated to any quality face criterion, the resulting polyhedra seem to be larger in volume. We can introduce the concept of terminal-face region as the union of the tetrahedra t whose $Lfpp(t)$, share a same terminal-face. Once an initial constrained tetrahedral mesh is available, further research questions arise: Which kind of polyhedra can be generated from terminal-face regions? Which quality face metric should be used to form terminal-face regions? Are polyhedral meshes based built from terminal-face regions composed of a lower number of elements, faces and edges than constrained Voronoi meshes with equivalent quality? Is it possible to design efficient point insertion strategies in order to refine polygonal/polyhedral meshes while fulfilling quality criteria? Is it possible to accelerate the generation of the initial polyhedral mesh as we did in 2D using the Gpu? Is it possible to accelerate the point insertion and optimization with Gpu computing?

The main research line of this project will be studying, developing and validating meshing algorithms based on terminal-edge and face regions for the generation of good quality polyhedral meshes for the VEM. In parallel we will continue working on the extension and improvements of the mixed-element mesh generator Mesher, based on modified octrees [42, 43], because it can be used as starting meshes for 3D groundwater modeling. Two research questions here are: i) What is the limit of pseudo-orthogonal edges for correct groundwater modeling? and ii) are only octree-based or Voronoi based meshes useful? Can more generic polyhedral elements be used? In our previous research projects, an extended Mesher has been built by integrating an algorithm to tessellate any convex polyhedron with Steiner points(1-irregular elements) into a Delaunay tessellation [18]. Mesher [19] is very robust, efficient and can generate meshes for very complex geometric domains. It can handle several surface triangulations defining the input geometry. Since the VEM accepts any polyhedron, we have been working on extending the original Mesher representing the domain geometry as it is specified in the input. Boundary polyhedra are formed by considering the intersection of input faces and the cut octant. Moreover, we are extending the refinement of octants to include the flexibility of the mesh generator [18, 28]. One of these is a flexible refinement approach into 2, 4 or 8 octants depending on the required refinement direction. We have been also working on a new metric for a reduce kind of polyhedral elements [20].

Related work

Mesh generation methods have been widely studied due to their importance in science and engineering applications, and in the computer graphics field, among others. Triangulations [44, 2, 1] or quadrilateral meshes [45, 46] are the most used with standard FEM in 2D. In 3D, tetrahedral meshes [42, 43, 47, 3, 48] or hexahedral meshes [49, 50] have been chosen depending on the problem to be solved. There are also some approaches to generate mixed element meshes [51, 28] and polygonal and polyhedral meshes [4, 52, 7, 31, 32, 33]. In these works polygonal/polyhedral cells are Voronoi regions. Different algorithms are described to generate constrained Voronoi meshes for specific applications. Meshing algorithms can be divided into two groups [53]: (1) direct algorithms: the discretization is generated from the input geometry, and (2) indirect algorithms: the discretization is generated starting from other mesh, commonly a triangulation. The polygonal/polyhedral meshing algorithms of this proposal are indirect methods except the ones based on modified octree approach.

The generalization of FEM to include polygons/polyhedra as part of the mesh elements was started in the last two decades [5, 6, 54]. Polygonal elements were generated by using a quadtree approach and from Voronoi cells. The VEM was introduced seven years ago [22, 23] and then several research groups have been developing computational frameworks for using the VEM, both in 2D and 3D, in order to explore how far the VEM can be used to solve new problems. In the last years, important results have been published related to the expected geometric quality of the polygonal/polyhedral cells to guarantee convergence and small errors using the VEM [55, 8, 30, 9]. These works also mention the need to count with refinement/improvement algorithms [56, 57] for generating a soft transition between finer and coarser regions of a polygonal/polyhedral mesh. To mention some applications, the VEM has been formulated and applied to solve linear elastic and inelastic solid mechanics problems [58], in fluid mechanics [59], in the optimization of a fluid problem through a discrete network [35], for compressible and incompressible nonlinear elasticity [60], for finite elasto-plastic deformations [61] and brittle crack propagation [34].

Only in the last decade, the size of basins and aquifers that need to be modeled have become regional (1000 km -10000 km). The large size of the domain area has led to hydrological modeling being carried out with unstructured grids [62, 26]. For this reason, Panday et al. [26] presented MODFLOW-USG, to support a wide variety of structured and unstructured meshes, including nested meshes and meshes based on prismatic triangles, rectangles, hexagons, and other cell shapes. The numerical method used by MODFLOW-USG is based on an underlying control volume finite difference (CVFD) formulation in which a cell can be connected to an arbitrary number of adjacent cells. For accurate solutions, the standard CVFD formulation requires that a line drawn between the centers of two connected cells intersects the shared face at a right angle. In our previous works for semiconductor devices simulations, we have been developing algorithms for the generation of polyhedral Delaunay meshes where the control volumes are the Voronoi regions [27, 28]. The meshes generated by VoroCrust could also be a possibility to explore [4]. Furthermore, the intersection point should coincide with an appropriate mean position on the shared face. This is a new criterion to take into account.

We are not going to be exhaustive reviewing parallel algorithms; we will mention mainly the ones related to gpu programming and meshing, that can be used as references in this project. On gpu, novel 2D parallel algorithms for the generation of Delaunay meshes are [63, 64, 65, 66]. To our knowledge the first implementation of Delaunay refinement using the GPU was presented in [67]. The 2D refinement algorithm is one order the magnitude faster than Triangle. The same group extended these ideas to implement 3D Delaunay refinement [68]. In [69] the authors present a set of recommendations to reduce waste threads in the GPU while implementing the Delaunay refinement algorithms in 2D and 3D. Since in this proposal, we are going to build polyhedra from terminal-face regions, and we plan to execute the refinement in the underlying tetrahedral mesh, it is worth to mention the parallel approaches for tetrahedral refinement using the Lepp concept. Rivara et al. [70] proposed a global parallel refinement for tetrahedral meshes based on the terminal edge refinement concept on a multi-core architecture. [71] presents an study of a multi-thread lepp-bisection algorithm for mesh refinement. For mesh improvement, [72] shows a parallel algorithm to untangle and to improve the quality of a mesh for distributed-memory machines and [73] proposes a parallel algorithm for implementing the edge-collapse strategy on gpu architectures.

Hypotheses

Our hypotheses are related mainly to the numerical methods that benefit from meshes composed of as less polyhedral cells as possible but respecting the point density requirements and the cell quality criteria. These are:

- Terminal-face regions can be transformed into simple polyhedral cells (convex or non convex) and the generated

partition built from these regions can be used as a polyhedral mesh

- Refinement/optimization strategies based on the Lepp concept can be extended to improve the quality of polygonal/polyhedral cells according to shape metrics required by the problem to be solved and the used numerical method
- Polyhedral meshes based in terminal-face regions will contain less points, edges and cells than constrained Voronoi meshes for the same point density requirements and similar accuracy in the simulations results
- Polyhedral meshes should allow us to perform more accurate and efficient hydrological modeling than structured meshes

General and specific goals

The main goal of this project is to design and develop new algorithms to generate quality polyhedral meshes that can be useful for modeling complex shapes and for solving scientific and engineering problems.

The specific goals to reach the general goal can be summarized as follows:

1. Identify and adapt quality face metrics to guide the union of tetrahedra to form polyhedral cells built from terminal-face regions. Study the geometric properties of polyhedral cells induced from terminal-face regions from both Delaunay and non-Delaunay partitions.
2. Design, implement and validate a sequential and parallel algorithm to generate an initial polyhedral mesh from terminal-face regions
3. Design, implement and validate refinement and optimization strategies for polyhedral cells to fulfill point density requirements and quality shape metrics imposed by application requirements
4. Compare the generated polyhedral meshes with constrained Voronoi meshes and modified octree based meshes in terms of number of points/elements, accuracy of the simulation results and simulation time/memory, among other comparison criteria.
5. Adapt/extend and evaluate the octree based and polyhedral meshes in the context of hydrological modeling, specifically groundwater flow simulations.

Methodology

In order to reach the general and specific goals described above, we plan to do the following:

1. Identify and adapt quality face metrics to guide the union of tetrahedra to form polyhedral cells built from terminal-face regions. Study the geometric properties of polyhedral cells induced from terminal-face regions from both Delaunay and non-Delaunay tetrahedral meshes.

During the development of the Polylla tool [15], we have shown that polygonal cells from Delaunay triangulations produce larger polygonal cells (and so less polygons) than the cells obtained from other triangulations. In 3D, the Delaunay mesh is not the one that maximizes the minimum solid angle [74], and the Lfpp properties have not been as studied as the Lepp properties [75, 76]. Therefore, we need to carry out a theoretical and empirical study in 3D to determine the size and simplicity of the terminal-face regions, among other properties, depending on the face metric to join tetrahedra. Since tetrahedra share triangles, we are going to study which 2D metric can help us to build quality polyhedral cells. Polyhedral cells could not be simple after this step as it happened in 2D. The triangle metrics can be area and edge-ratio, among others. Here the concepts of terminal-face, frontier-face and internal-face have to be formalized. We plan to design and run several experiments to validate the algorithm and obtain empirical results about the quality of the generated polyhedra. We are going to use as a reference for the quality criteria expected for polygons/polyhedra the study done recently by Sorgente et al. [8, 30, 9] to guarantee convergence for the VEM. The geometric properties described in these previous works will be used to study the shape of the polyhedral cells built from terminal face regions.

2. Develop a sequential and parallel algorithm to generate an initial polyhedral mesh from terminal-face regions

During this year we have been studying the Lepp in 3D to generate 3D terminal-edge regions. Preliminary results show that the generated cells are not simple and composed of few tetrahedra. Small tests with terminal-face regions seem to generate better results. For this specific goal, we trust in robust tools to generate constrained tetrahedral meshes with the required mesh density such as TetGen [3] or TetWild [48].

In 2D, we have developed a prototype version of a parallel Polylla using the Gpu [16]. This parallel algorithm was inspired in the ideas for parallel edge classification while identifying terminal edge regions in [77]. We will explore how far we can go to parallelize some steps of the 3D algorithm. As in 2D, we expect that the generation of the polyhedral mesh from terminal face regions should be accelerated using the Gpu. We have found that some steps are data parallel and others task parallel. We would like to explore if the mix of both strategies can accelerate most.

3. Design, implement and validate refinement and optimization strategies for polyhedral cells to fulfill point density requirements and quality shape metrics imposed by application requirements

It is very difficult to estimate the right point density from scratch. Moreover, adaptive refinement requires to count with algorithms to split elements that are still too large or to add additional points in some regions of the domain. That is why a useful meshing tool should provide algorithms to refine a mesh by adding new points and/or optimizing the shape of elements by moving points among others. Our 2D polygonal meshing tool Polylla still does not include refinement/optimization algorithms but our current work includes the implementation of algorithms to i) transform a Polylla mesh with convex and not convex cells to a Polylla mesh with only convex cells and ii) dividing polygonal elements to fulfill area requirements or a higher point density in some region of interest (ROI). Our approach is to keep the underlying triangulation associated to each polygonal cell because (i) a non-convex polygon can be transformed into convex polygons choosing one or more triangulation edges and (ii) insertion points strategies available for the Lepp can be adapted and applied to the involved triangles. From the new triangulation, new polygonal cells are built.

To fulfill this specific goal, we will first validate our 2D approach. Preliminary results are promising but we need still improve the transition between coarser and finer regions of the domain. We think that in 3D we should explore first a similar strategy. Refining polyhedral elements is quite expensive, then the use of the underlying tetrahedral mesh to increase the point density/refine the tetrahedra, and then build the new polyhedral cells from the refined tetrahedral mesh seems a good alternative.

4. Compare the generated polyhedral meshes with constrained Voronoi meshes in terms of number of points/elements, accuracy of the simulation results and simulation time/memory, among other comparison criteria.

We will constantly review the literature to select the most competitive algorithms to compare our work. Currently we can mention tools like VoroCrust [4] to compare our polyhedral meshes against constrained Voronoi based meshes in 3D. We plan to compare the 3D polyhedral meshes as we did in 2D with the polygonal meshes [15]. Our strategy was by starting from the same input Delaunay triangulation, with the required point density, the final polygonal meshes require less cells, edges and points to discretize properly the domain than the constrained Voronoi meshes. We are aware that this comparison might be not fully fair because the Voronoi mesh is generated from the Delaunay triangulation and not from scratch. To our knowledge, we have not seen any approach to generate as much Voronoi seeds (input points) so that the computed Voronoi points fulfill the density requirements with a minimum number of points. Another issue with Voronoi based meshes is to generate constrained Voronoi meshes. The available tools such as Detri2d, cut the boundary Voronoi cells from a conforming Delaunay mesh, adding new points to the boundary edges. Approaches like VoroCrust [4], in 3D, place carefully Voronoi seeds across the domain surface to avoid the need of clipping. These new points increase the number of final points too. Moreover, since during the mesh refinement and improvement steps, the algorithm should always consider the possibility of generating non-convex polygons if they are adequate and so keep the number of inserted points as low as possible, we will evaluate how strong can be the performance improvement in both storage and computational time while generating the mesh and solving specific problems with the VEM.

For visual comparison, we have developed the visualization tool called Camarón [78] for the inspection of polygon and polyhedral meshes, manifold and non-manifold geometries. Several geometric quality criteria for both surface and volume meshes are already implemented, and also contour lines and isosurfaces can be built from scalar simulation data. We plan to add new functionalities that allow us to visualize the quality criteria related to the VEM.

- Adapt/extend and evaluate the octree based and polyhedral meshes in the context of hydrological modeling, specifically groundwater flow simulations.

We will review most popular methods applied to model aquifers with scarcity problems and test theoretical cases inspired in real aquifers. The test cases will be oriented to be scaled to real problems in the future. The most classical problems have been solved with structured grids in the past, however, currently the construction of unstructured grids [26] would allow to address regional scale problems without losing resolution with a much smaller number of elements (called HU, Hydrogeological Units). First we will extend Mesher to create 3D meshes whose elements fulfil geometrical requirements for numerical modeling with MODFLOW-USG, and then we will carry out simulations with theoretical cases. We hope to be able to model real cases at the end of the project.

The chosen programming language chosen is c++ because we can program very efficiently and at the same time use the object oriented paradigm when is appropriate. For parallel programming we have experience with CUDA and OPENCL. The advantage of OPENCL is that we can use any video card while cuda can be only used with NVIDIA cards. As design and programming methodology we are using an object oriented approach which has allowed us to build software easy to extend, adapt and reuse. We put emphasis in building reusable and extensible components, where efficiency is not sacrificed.

Work Plan

Depending on the interest of the students, some activities described in Table 1 can be interchanged. Each year, we plan to send at least one paper to an international journal (Engineering with computers, CAD journal and International journal of numerical Methods and Engineering) and at least one paper to an international conference (International Meshing Roundtable IMR, Grapp conference or EuroCG workshop, among others).

Table 1: Work Plan

Goal	Year 1	Year 2	Year 3	Year 4
Identify, adapt face metrics. Study geometric properties of the induced polyhedra.	Review and select face metrics. Implement and test the metrics on the resulting cells	Study the properties of the resulting cells, from Delaunay tetrahedral meshes	Study the properties of the resulting cells from non-Delaunay tetrahedral meshes	Evaluation and theoretical formalization
Develop sequential and parallel algorithms to generate polyhedral cells	Improve sequential algorithms to generate polyhedral meshes from terminal-face regions. VEM validation in small tests	Develop a gpu parallel algorithm to generate polyhedral meshes from terminal-face regions	Performance evaluation between both approaches. Improvements	Study the scalability of the algorithms. Design experiments
Design, implement and validate refinement and optimization strategies	Review point insertion algorithms in tetrahedra. Evaluation in small cases	Include a first refinement guided by quality metrics. Optimization of of polygons/polyhedra	Include refinement in regions of interest. Ensure soft transition between refinement levels	Include other refinement/improvement strategies
Compare the generated polyhedral meshes with constrained Voronoi meshes in terms of several metrics	Add metrics for the evaluation and visualization of cells in Camaron	Comparison in terms of the number of points, cells and computational cost. Comparison with respect to the quality of the mesh cells	Global evaluation of the generated meshes. Analysis and comparison of simulation results	Test with the VEM. Impact analysis
Adapt/extend and evaluate polyhedral meshes in the context of hydrological modeling.	Review methods to model aquifers. Test theoretical cases. Study mesh requirements	Extend our modified octree approach and evaluate its use with MODFLOW-USG	Study the impact of the (non)orthogonality criterion in groundwater modeling	Test the use of generic polyhedra. Design future work

Work in progress

As stated in the formulation of this project, this proposal is the continuation of our current Fondecyt project *N°* 1211484, which finishes in March 2024. For the validation of the proposed polyhedral meshes to solve problems with the VEM, we are collaborating with Dr. Alejandro Ortiz-Bernardin from the Mechanical Engineering Department at the University of Chile [14, 79, 80]. The student Sergio Salinas is finishing his PhD (granted by our science foundation) in this topic under my supervision. During the last three years, together with theoretical formulations, he has been the main developer of three tools: the polygonal mesh generator Polylla (<https://github.com/ssalinasfe/Polylla-Mesh-DCEL>) [15], the compact version of Polylla (<https://github.com/ssalinasfe/Compact-Polylla-Mesh>) [17] and The gpu version of Polylla (<https://github.com/ssalinasfe/GPolylla>) [16]. Preliminary versions in python for the study of the generation of polyhedral cells from terminal-edge and terminal-face regions are available in <https://github.com/ssalinasfe/3D-Polylla>. Currently, the student Magdalena Alvarez is starting her thesis (coguided by Sergio Salinas) to study the kind of polyhedra generated from these regions and to optimize the code. The co-researcher Claudio Lobos will continue working in adding flexible refinement to Mesher (<https://github.com/clobosmesher/MixedOcTree>), the modified oc-tree mesh generator, and determining for which kind of problems these meshes can be used with the VEM. Another aspect will be to add new data structures for faster mesh generation and also to focus on element quality metrics to count with mesh repairing techniques. These advances will also be included in the quadtree version of Mesher (<https://github.com/clobosmesher/MixedQuadTree>). The main goal of these improvements will be to produce a better version of Mesher to be used in the context of groundwater modeling with the collaboration of the other co-researcher Dr. Pedro Sanzana. It is worth to mention that Pedro Sanzana and me have been collaborating in polygonal meshes for hydrological modeling in catchments for almost 10 years. In the last three years we have also collaborated in a real case study of water resources in the Azapa Valley, mixing aquifers and superficial sources granted by the chilean government. A new PhD student, Roberto Carrasco, granted by our science foundation, has already collaborated in GPolylla and will continue collaborating in the design of Gpu algorithms.

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In this section, include the list of cited references in the Proposed Research section. **Length: 5 pages.**
(Verdana font size 10, letter size is suggested).

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